

Cross-city path-loss transfer for digital-twin network planning: why tile parameters fall short and how site-aware link models close the gap

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Abstract

Digital-twin network planning needs path-loss models for areas without measurements. Tile-based methods fit a log-distance exponent and intercept per map tile and transfer them to unmeasured tiles by descriptor similarity. We study this transfer across cities on reproducible ray-traced corpora: 11 geo-referenced cities with 220 rooftop sites at 3.5 GHz (8,977,400 links) and at 7.5 GHz, a random-siting replicate and independently modelled external scenes. We evaluate transfer where a planner uses it: path loss, serving site, SINR, coverage, cell edges and site planning. The tile labels are imprecise: with a small-sample cluster-robust variance their uncertainty is 3.2 times the naive one, and they change with the serving sites and with the dynamic range of the data. Learned regressors transfer them better than the source median, and learning the metric of Gaussian-kernel transfer removes most of its disadvantage, but even perfect tile labels give a path-loss error of 11.6 dB and the correct serving site for only 50.1% of the pixels. A site-aware hurdle model that predicts each link, and whether it exists, from its obstruction and diffraction geometry reduces the error on unseen cities to 9.3 dB, raises serving-site accuracy to 68.8% and cuts the coverage lost when planning five sites from 20.6% to 0.5%, in every held-out city. A city-level coverage certificate lets it promise 92.0% of the pixels at a guaranteed risk, against 78.3% for tile transfer. Data, code and every result are released and regenerated by one command.

Keywords: digital twin, network planning, path-loss modelling, transfer learning, ray tracing, coverage certification, empirical Bayes

1. Introduction

Network planning with a digital twin requires a propagation model for places where no measurement has been taken yet: coverage, interference and site selection must be assessed before a campaign exists in the target region [1, 2]. Ray tracing resolves the geometry but must be rerun for every change of layout and needs detailed 3-D data [3, 4]; empirical formulas such as Okumura–Hata or 3GPP TR 38.901 are cheap but apply one parameter set to a whole environment class [5, 6]. Tile-based models sit between the two: the twin is partitioned into tiles, each tile carries the two parameters (γ, P_{L_0}) of a log-distance law, and parameters learned where measurements exist are transferred to unmeasured tiles by environmental similarity [7, 8]. The transported quantity is small and physically interpretable, and a planning tool can evaluate it for any site layout.

Whether such transfer works *across cities*, and what its errors cost a network planner, has not been established. The

descriptor-based framework of [8], which transfers tile parameters with a Gaussian kernel over a descriptor space, was evaluated on two urban datasets that are not public. Three questions remain open. First, how reliable are the tile labels themselves, given that a tile of 100 m sees its serving sites over a narrow range of distances and that neighbouring receivers share propagation paths? Second, how does kernel transfer behave when the target city differs from every source city, which is exactly when transfer is needed, and how does it compare with learned regressors under one protocol? Third, what do parameter errors mean for the quantities a planner uses: path loss, serving cell, SINR, coverage, cell edges and the choice of sites, is the tile parameter the right quantity to transfer at all, and how much coverage can an operator promise from a transferred model?

This paper answers these questions on a public benchmark. Our contributions are the following.

1. **Reproducible multi-city, multi-band corpora.** Eleven geo-referenced city scenes built from a pinned release of open building data are ray traced from 220 rooftop sites at 3.5 GHz (8,977,400 links, 10,428 labelled tiles) and at 7.5 GHz in the upper mid-band, with a ray-budget study, a physical consistency check of the solver output, a replicate with random rooftop siting and an external test on independently modelled scenes. Every number in this paper is

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regenerated from the released data by one command.

2. **Tile labels with honest precision.** We show that the per-tile fit is ill-conditioned (Proposition 3), replace ad-hoc regularisation by empirical-Bayes shrinkage whose risk is bounded whatever the tile geometry (Proposition 4) with a small-sample (CR2) cluster-robust variance, and show that the labels depend on the serving sites, the tile size and the dynamic range of the data.
3. **An analysis of kernel transfer.** We prove that the Gaussian kernel operator has two limiting regimes (Theorem 1), measure where the tuned operator lies on every held-out city, and show that learning its metric removes most of its disadvantage.
4. **A leave-one-city-out benchmark with city-level inference.** Standard-model, label-averaging, Gaussian-process, linear, tree, neural and domain-adaptation operators are compared under one tuning protocol with five seeds, with and without surveyed tiles (random, contiguous or designed), and every claim is tested at the level of cities with bootstrap intervals, Wilcoxon tests and Holm correction.
5. **Network-level evaluation and site-aware link transfer.** We derive what a tile model can express for cell association and SINR (Proposition 1), evaluate every operator through path loss, serving cell, SINR, coverage, cell edges, dimensioning and site planning scored on the ray-traced truth, and propose a hurdle link model conditioned on the obstruction geometry of each link that outperforms even the tile model with perfect labels in every held-out city.
6. **Survey design and coverage certificates.** We give a survey design for MDT-style site calibration with a $(1 - 1/e)$ guarantee (Proposition 5) and a distribution-free, city-level coverage certificate from leave-one-city-out residuals (Proposition 6), and measure what each transfer model can promise at a stated risk.

Section 2 reviews related work, Section 3 formalises the network model and the transfer problem, Section 4 describes the corpus, labels, descriptors, operators and evaluation, Section 5 reports the results, Section 6 discusses their implications and limitations, and Section 7 concludes. Proofs are in Appendix A and the complete city-level comparisons in Appendix C.

2. Related work

Empirical and site-specific propagation models. Empirical models such as Okumura–Hata [5] and the 3GPP TR 38.901 formulas [6] express path loss through a small number of parameters fitted to measurement campaigns and applied uniformly over an environment class. Measurement studies showed early that the path-loss exponent is itself a random quantity that changes from cell to cell [9], which is the variability a tile-level model tries to capture. Deterministic ray tracing resolves the geometry explicitly [3]; differentiable ray tracers such as Sionna RT make city-scale simulation and calibration against measurements practical [4, 10]. We use ray tracing to produce ground truth for many cities, not as the run-time predictor.

Learned radio maps and path-loss regression. Convolutional [11], autoencoder [12], adversarial [13] and diffusion [14] models predict dense radio maps from environment maps, and radio-map estimation from sparse measurements has a mature statistical treatment [15]. Channel knowledge maps [16] generalise the idea to other channel quantities. Regression of path loss from engineered or image features has been studied with measurements at 2.6GHz [17] and 28 GHz [18], across several cities with obstruction-height maps [19], and is surveyed in [20]. These methods predict path loss or maps directly; the tile-parameter approach studied here predicts a compact, physically interpretable pair per tile that a planning tool can evaluate for any site layout.

Tile-based transfer and digital twins. Digital twins of wireless networks [1, 2] need propagation models for regions where no measurements exist yet. MAPLE divides 3-D city maps into tiles and estimates mixed-path propagation in a new deployment area from the similarity of its tiles to tiles of measured areas [7]. The descriptor-based framework of [8] describes each tile by terrain, built-structure, land-cover and climatic descriptors, calibrates log-distance parameters per tile from measurements, and transfers them to unmeasured tiles by a Gaussian-kernel weighting of descriptor similarity, with a cross-city evaluation on two urban datasets. We study the same transfer problem on a larger public corpus, analyse the kernel operator, compare it with learned regressors under a common protocol, and evaluate the consequences for network planning.

Transfer under covariate shift and planning. Transfer between domains with different input distributions is covariate shift [21], and its difficulty depends on the discrepancy between the domains [22]; importance weighting [21] and the alignment of feature statistics [23] are standard remedies, which we include as baselines. In operational networks, targeted measurements can be collected with the minimisation-of-drive-tests (MDT) framework [24, 25], which motivates our few-shot setting; choosing where to measure is a submodular sensor-placement problem [26]. Site selection with propagation predictions is a classical optimisation problem [27]; coverage maximisation is submodular and admits greedy approximation guarantees [28].

Uncertainty and censored measurements. Conformal prediction gives distribution-free prediction sets under exchangeability [29], and leave-one-out variants such as the jackknife+ reuse every training point for calibration [30]; we apply the exchangeability argument at the level of cities, the unit of shift. Measurements with a detection floor are censored, for which the Tobit model is the classical treatment [31]; received-power data beyond a receiver’s sensitivity are censored in exactly this sense.

3. System model and problem formulation

3.1. Network and propagation model

A city contains a set $\mathcal{B}$ of base-station sites and a set $\mathcal{P}$ of receiver locations (pixels) at user height. The link (p, b) has three-dimensional length d_{pb} and path loss $P_L(p, b)$ in dB. With trans-

mit power P_{tx} (dBm) at every site and noise power N , the received power, serving site and downlink signal-to-interference-plus-noise ratio under full load are

$$P_{pb} = P_{\text{tx}} - P_L(p, b), \quad s(p) = \arg \max_{b \in \mathcal{B}} P_{pb},$$

$$\text{SINR}(p) = \frac{10^{P_{ps(p)}/10}}{\sum_{b \neq s(p)} 10^{P_{pb}/10} + 10^{N/10}}. \quad (1)$$

A pixel is *covered* at coupling-loss level L if $P_L(p, s(p)) \leq L$, and its spectral efficiency is obtained from the SINR with the attenuated and truncated Shannon mapping used in 3GPP system evaluations [32].

The city is partitioned into square tiles of side G_s . Inside tile t the path loss follows the log-distance law [33, 9]

$$P_L(p, b) = P_{L_{0,t}} + 10 \gamma_t \log_{10} \frac{d_{pb}}{d_0} + \varepsilon_{pb}, \quad p \in t, \quad (2)$$

with exponent γ_t , intercept $P_{L_{0,t}}$ at the reference distance d_0 and a residual ε_{pb} of variance σ_{ε}^2 . We use $d_0 = 100$ m, the tile scale, so that $P_{L_{0,t}}$ is an interpolated rather than an extrapolated quantity; the intercept at 1 m follows as $P_{L_{0,t}} - 20\gamma_t$. Writing $x = 10 \log_{10}(d/d_0)$, Eq. (2) is linear in $\theta_t = (\gamma_t, P_{L_{0,t}})^T$ with design matrix $\mathbf{A}_t = [\mathbf{x} \ \mathbf{1}]$.

Equation (2) is the interface between the digital twin and network planning: a planning tool that knows θ_t for every tile and the site positions can evaluate Eq. (1) anywhere without ray tracing. Two structural consequences of the tile model are used later.

Proposition 1 (What a tile model can express). *Let every link seen from pixel p use the parameters of the tile containing p , with $\gamma_t > 0$. Then (i) the predicted serving site is the nearest site, $s(p) = \arg \min_b d_{pb}$, whatever θ_t ; and (ii) the predicted SINR is*

$$\text{SINR}(p) = \frac{(d_{ps}/d_0)^{-\gamma_t}}{\sum_{b \neq s} (d_{pb}/d_0)^{-\gamma_t} + 10^{(N - P_{\text{tx}} + P_{L_{0,t}})/10}}, \quad (3)$$

so in the interference-limited regime it depends on the exponent only, while coverage at level L depends on both parameters.

The proof is immediate from Eqs. (1) and (2). Part (i) bounds what any tile-parameter predictor can achieve in cell association: the metric is identical for every transfer operator and measures the tile model itself. Part (ii) identifies which parameter errors matter for which planning quantity.

3.2. The transfer problem

Each tile carries a descriptor $\mathbf{v}_t \in \mathbb{R}^p$ computed from the geometry of the digital twin. Measurements, and hence labels θ_t , are available in a set of *source* cities $\mathcal{T}_S = \{(\mathbf{v}_j, \theta_j)\}_{j=1}^N$ and not in the *target* city, whose tiles have descriptors only. A *transfer operator* maps the source data and a target descriptor to a parameter estimate, $\hat{\theta}_u = \mathcal{A}(\mathcal{T}_S; \mathbf{v}_u)$. Source and target descriptors follow different distributions: the purpose of transfer is to serve cities unlike those measured, so covariate shift [21, 22] is the

defining feature of the problem. In the *few-shot* variant, k target tiles are surveyed and their labels revealed before prediction.

The operator is judged at two levels: the parameter error on the target tiles and, more importantly for planning, the error of the network quantities of Section 3.1 computed from the predicted parameters and compared with the ray-traced network.

3.3. Risk of label-averaging operators

Many transfer operators, including the Gaussian-kernel operator of [8] and k -nearest-neighbour averaging, predict a convex combination of source labels, $\hat{\theta}_u = \sum_j w_j(\mathbf{v}_u) \theta_j$ with $w_j \geq 0$, $\sum_j w_j = 1$ and weights that depend on descriptors only.

Proposition 2 (Risk decomposition). *Let the source labels be $\theta_j = \theta^*(\mathbf{v}_j) + \eta_j$, with $\theta^*(\mathbf{v})$ the conditional mean parameter and η_j zero-mean label noise of covariance Σ_η , independent across j and of the descriptors. For a label-averaging operator the target risk $R = \mathbb{E}_{\mathbf{v}} \|\hat{\theta}(\mathbf{v}) - \theta^*(\mathbf{v})\|^2$ is*

$$R = \underbrace{\mathbb{E}_{\mathbf{v}} \left\| \sum_j w_j(\mathbf{v}) \theta^*(\mathbf{v}_j) - \theta^*(\mathbf{v}) \right\|^2}_{\text{smoothing bias}} + \underbrace{\text{tr}(\Sigma_\eta) \mathbb{E}_{\mathbf{v}} \sum_j w_j(\mathbf{v})^2}_{\text{noise transmission}}. \quad (4)$$

The second term equals $\text{tr}(\Sigma_\eta) \mathbb{E}[1/n_{\text{eff}}]$ with the effective neighbourhood size $n_{\text{eff}}(\mathbf{v}) = 1/\sum_j w_j(\mathbf{v})^2$. Concentrated weights lower the bias and raise the noise transmission; diffuse weights do the opposite. Under covariate shift the bias term grows, because the source set may contain no tile whose conditional mean resembles the target's; with noisy labels the second term matters as well. Ridge regression is also linear in the labels, and so is a regression tree once its partition is fixed; the difference is that their weights are not confined to the probability simplex and are shaped by the descriptor-label relation rather than fixed by an isotropic distance, which is the distinction that matters below.

4. Methods

Figure 1 summarises the pipeline. Every stage is implemented in the accompanying repository and executed by a single command; every number reported in Sections 5–6 is generated from its outputs.

4.1. Ray-traced corpus

Each city is a square of side 3 km centred on a fixed coordinate and expressed in a local metric frame, the UTM projection of its zone shifted to the centre [34]. Building footprints are taken from the buildings theme of Overture Maps [35], release 2026-09-23.0, a versioned open dataset whose primary source is OpenStreetMap [36]; a pinned release makes the geometry exactly reproducible, which a live map query is not. Because grid north and true north differ, the footprint query box covers all four corners of the square. Building heights are resolved in order from the height attribute, from the number of floors at 3.2 m per floor, and otherwise from the median of the resolved heights of the city; the share of each source is reported per city. Footprint exteriors are clipped to the square, parts smaller

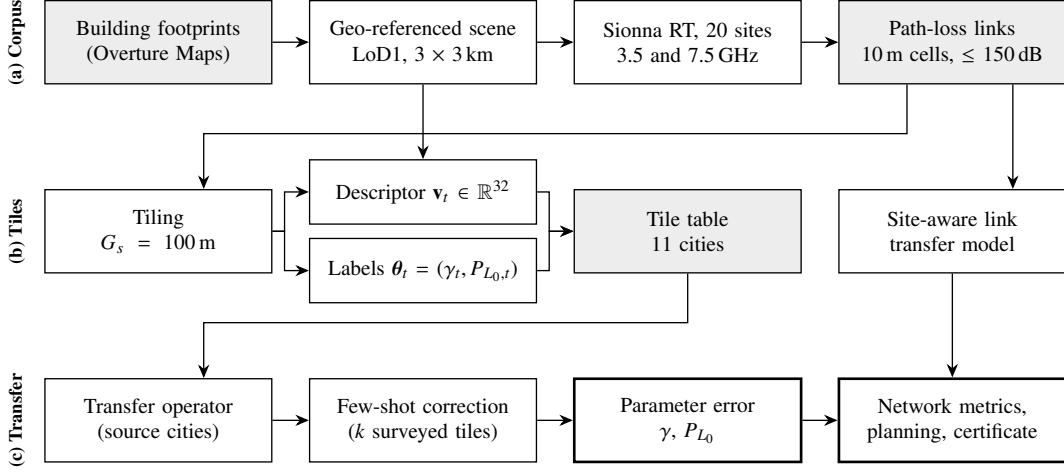


Figure 1: Processing pipeline. (a) Geo-referenced scenes are built from open building footprints and ray traced. (b) Links are grouped into tiles; each tile receives a label $\theta_t = (\gamma_t, P_{L_0,t})$ and a descriptor $\mathbf{v}_t \in \mathbb{R}^{32}$ computed from the same scene geometry. (c) Tile transfer operators fitted on source cities predict the labels of a held-out city, optionally corrected with k surveyed tiles; the site-aware link model predicts every link from its geometry. Both are evaluated through the network quantities of Section 3.1, a site plan and a coverage certificate.

than 4 m^2 are discarded, and the remainder is extruded to level-of-detail 1 and placed on a flat ground plane. Buildings use the ITU concrete model and the ground the ITU medium-dry-ground model [37].

Each city has 20 sites on rooftops: local height maxima of the building mesh on a 25 m grid are ranked by height and selected greedily with a minimum separation of 250 m , and the antenna is mounted 6 m above the roof; if too few candidates satisfy the separation, the remaining sites are drawn at random from the other rooftop candidates with a fixed seed. Radio maps are computed with Sionna RT [4, 10] (version 2.1.0) at 3.5 GHz with isotropic antennas, 1×10^8 rays per site, maximum interaction depth 5, specular and diffuse reflection, refraction and diffraction (including edge diffraction) enabled, on a 10 m grid at 1.5 m receiver height. Cells with path loss above 150 dB , beyond any practical coupling loss, are discarded. The solver version and settings are stored with every city. Twenty sites on 9 km^2 give one site per 0.45 km^2 , the density of a hexagonal layout with an inter-site distance of about 720 m , sparser than the 500 m of the 3GPP urban-macro scenario [6]; ranking rooftops by height follows macro-cell practice. Both choices are tested in Section 5.9.

Two further corpora use the same scenes and settings. The *second band* re-traces every site at 7.5 GHz , in the 7 GHz to 24 GHz range studied for NR [38]. The *siting replicate* re-traces 4 cities (Amsterdam, London, New York, Paris) at 3.5 GHz with the same number of sites taken from the rooftop candidates in random order instead of by height, under the same separation rule.

Consistency check. The planar radio-map estimator of the solver weights a ray crossing the measurement plane by the inverse cosine of its incidence angle, and a diffracted ray by an integration weight over the diffraction cone; both weights are unbounded, so a cell occasionally receives one overweighted contribution. Its path loss can then fall below free space by

tens of decibels, which no incoherent sum of a few propagation paths allows. We remove every link whose path loss lies more than 6 dB below the free-space loss at its 3-D distance: $1,170$ links, 0.013% of the corpus. For the site with the most such links, disabling diffraction reduces the links more than 10 dB below free space from to with otherwise identical settings, which identifies the diffraction weights as their source.

4.2. Tile labels

Links are assigned to tiles of side $G_s = 100 \text{ m}$ by their receiver position; tiles with fewer than 30 links are not labelled. Inside a tile, the ordinary least-squares (OLS) slope is poorly determined because every receiver sees the serving sites from a similar distance.

Proposition 3 (Ill-conditioning of the per-tile fit). *For $\mathbf{A} = [\mathbf{x} \ \mathbf{1}]$ with n rows, $\det(\mathbf{A}^\top \mathbf{A}) = n^2 \text{Var}(x)$ and $\text{Var}(\hat{\gamma}_{\text{OLS}}) = \sigma_\varepsilon^2 / (n \text{Var}(x))$, which diverges as the intra-tile spread of x vanishes.*

A receiver at distance d from its site sees a tile of side G_s span only $\Delta x \approx 10 \log_{10}(1 + G_s/d)$, which is 0.41 dB for $G_s = 100 \text{ m}$ and $d = 1 \text{ km}$. We therefore shrink the slope towards the city-wide mean with weights set by the data (empirical Bayes [39]).

Proposition 4 (Empirical-Bayes shrinkage). *Let b_t be the OLS slope of tile t , computed with x centred inside the tile, with sampling variance v_t , and let the true slopes vary between tiles with mean μ and variance τ^2 . Among estimators $\hat{\gamma}_t = (1 - w)b_t + w\mu$ the mean squared error $(1 - w)^2 v_t + w^2 \tau^2$ is minimised by*

$$w_t = \frac{v_t}{v_t + \tau^2} = \frac{\lambda_t}{\lambda_t + S_{xx,t}}, \quad \lambda_t = \frac{S_{xx,t} v_t}{\tau^2}, \quad (5)$$

which is ridge regression on the centred slope with penalty λ_t and prior mean μ . Its risk is $v_t \tau^2 / (v_t + \tau^2) \leq \min(v_t, \tau^2)$, bounded by τ^2 however degenerate the tile geometry.

Algorithm 1: Tile labels

Input: links (x_i, y_i, t_i, b_i) of one city; minimum count $n_{\min}$

Output: $\hat{\theta}_t, w_t$, iid and cluster-robust standard errors

```

1 foreach tile  $t$  with  $n_t \geq n_{\min}$  do
2    $\bar{x}_t, \bar{y}_t, S_{xx,t}, S_{xy,t}; \quad b_t \leftarrow S_{xy,t}/S_{xx,t}; \quad v_t \leftarrow$ 
   cluster-robust variance of  $b_t$  (Eq. (6))
3 end
4  $(\hat{\mu}, \hat{\tau}^2) \leftarrow$  DerSimonian–Laird over  $\{(b_t, v_t)\}$ 
5 foreach tile  $t$  do
6    $w_t \leftarrow v_t/(v_t + \hat{\tau}^2); \quad \hat{\gamma}_t \leftarrow (1 - w_t)b_t + w_t\hat{\mu};$ 
    $\hat{P}_{L_0,t} \leftarrow \bar{y}_t - \hat{\gamma}_t\bar{x}_t$ 
7   residuals  $e_i \leftarrow y_i - \hat{P}_{L_0,t} - \hat{\gamma}_tx_i$ 
8   standard errors from Eq. (6) with iid and with site
   clusters
9 end

```

Radio-map cells served by the same site share propagation paths, so the independent-residual variance $s_t^2/S_{xx,t}$ understates v_t . We therefore use the cluster-robust variance of Eq. (6) (below) of the OLS slope, with the serving sites as clusters, never below the independent value; a tile served by a single site uses the independent value multiplied by the median ratio of the two over the city (its design effect). We estimate (μ, τ^2) per city with the DerSimonian–Laird moment estimator [40] over the tiles with a non-degenerate span, set $\hat{\gamma}_t = (1 - w_t)b_t + w_t\hat{\mu}$, and leave the intercept unpenalised: $\hat{P}_{L_0,t} = \bar{y}_t - \hat{\gamma}_t\bar{x}_t$. A tile with no span takes $\hat{\gamma}_t = \hat{\mu}$. The shrinkage weight w_t is published with every label. Algorithm 1 states the procedure; it is vectorised over all links of a city.

Label precision. The estimator is linear in the path losses. In the centred parametrisation $y_i = a_t + \gamma_t(x_i - \bar{x}_t) + e_i$ with design rows $\mathbf{z}_i = (1, x_i - \bar{x}_t)$, the coefficients are $\mathbf{M} \sum_i \mathbf{z}_i y_i$ up to the prior term, with $\mathbf{M} = \text{diag}(1/n, 1/(S_{xx,t} + \lambda_t))$ ($\lambda_t = 0$ for the OLS slope), and $\hat{P}_{L_0,t} = \hat{a}_t - \hat{\gamma}_t\bar{x}_t$. With the links of the tile partitioned by serving site into G clusters $\mathcal{I}_1, \dots, \mathcal{I}_G$, with design blocks $\mathbf{Z}_c$ and residuals $\mathbf{e}_c$, the cluster-robust variance is

$$\widehat{\text{Var}}_{\text{cl}} = \mathbf{M} \left(\sum_{c=1}^G \mathbf{u}_c \mathbf{u}_c^\top \right) \mathbf{M}, \quad \mathbf{u}_c = \mathbf{Z}_c^\top (\mathbf{I} - \mathbf{H}_{cc})^{-1/2} \mathbf{e}_c, \quad (6)$$

with $\mathbf{H}_{cc} = \mathbf{Z}_c \mathbf{M} \mathbf{Z}_c^\top$ (Moore–Penrose inverse root where $\mathbf{I} - \mathbf{H}_{cc}$ is singular). This is the bias-reduced linearisation (CR2) of [41, 42]; without the factor $(\mathbf{I} - \mathbf{H}_{cc})^{-1/2}$ it is the usual estimator [43, 44], which is biased downwards when the clusters are few, as here. Since $\mathbf{Z}_c^\top f(\mathbf{Z}_c \mathbf{M} \mathbf{Z}_c^\top) = f(\mathbf{Z}_c^\top \mathbf{Z}_c \mathbf{M}) \mathbf{Z}_c^\top$ for $f(s) = (1 - s)^{-1/2}$, every $\mathbf{u}_c$ requires only the eigen-decomposition of a 2×2 matrix. The iid variance is $s^2 \mathbf{M} (\sum_i \mathbf{z}_i \mathbf{z}_i^\top) \mathbf{M}$. Because the number of clusters per tile is small, Eq. (6) is complemented by a split-half estimate: the sites of a city are divided at random into two disjoint halves, the complete estimator (including its own $\hat{\mu}, \hat{\tau}^2$) is refitted on each half, and $\text{sd}(\hat{\gamma}^{(1)} - \hat{\gamma}^{(2)})/\sqrt{2}$ estimates the standard error of a half-data label; dividing by a further $\sqrt{2}$ extrapolates to the full data under the assumption

that the variance scales with the inverse number of sites. Five random partitions are pooled.

4.3. Tile descriptors

The descriptor must describe exactly the geometry the ray tracer saw. It is therefore computed from the same footprint parts, heights and scene clipping as the extruded mesh, with every part clipped to the tile it falls in. It contains the 32 built-environment features of [8]: footprint ratio and building count in five height classes ($< 5, 5-10, 10-30, 30-60, \geq 60$ m); mean and standard deviation of footprint area, height and volume; total footprint density; and, for horizontal planes at 5, 10, 30, 60 and 100 m, the number of buildings reaching the plane and the mean and standard deviation of the distance between their centroids. The topographic, land-cover and climatic descriptors of [8] are omitted on purpose: the scenes have a flat ground, no water or vegetation surfaces and no weather, so these features would describe nothing the solver modelled and could only act as city identifiers. Descriptors are stored unnormalised; every experiment scales them by the column-wise maximum absolute value of its training cities, never of the target.

4.4. Transfer operators

We compare constant predictors, label-averaging operators and learned regressors.

- *Source median:* the median source label, the no-information reference.
- *3GPP UMa:* the line-of-sight and non-line-of-sight single-slope formulas of TR 38.901 [6] at the reference distance, which give fixed parameters for every tile, and *COST-231 Hata* [45] (metropolitan correction, median site height of the source cities), used outside its validity range of 1.5 GHz to 2 GHz and 1 km to 20 km as a further standard model.
- *Kernel (Nadaraya–Watson)* [46, 47], the operator of [8]:

$$\hat{\theta}_u = \sum_j \tilde{w}_j(u) \theta_j, \quad \tilde{w}_j(u) \propto \exp\left(-\frac{\|\mathbf{v}_u - \mathbf{v}_j\|^2}{2\sigma^2}\right), \quad (7)$$

computed with the minimum squared distance subtracted in the exponent, which leaves the weights unchanged and avoids numerical underflow.

- *GP-ARD:* a Gaussian process with one length scale per descriptor (automatic relevance determination) [48], i.e. a kernel operator whose metric is learned; the hyperparameters maximise the marginal likelihood on 600 random source tiles and the posterior mean uses every source tile.
- *k-NN:* the mean label of the k nearest source tiles.
- *Ridge,* a two-layer *MLP*, gradient-boosted trees (*GBDT*, *LightGBM* [49]) and a *neural encoder* with a constrained (γ, P_{L_0}) head trained with a heteroscedastic Gaussian likelihood.

- Two domain-adaptation variants that use the unlabelled descriptors of the target city: *GBDT-IW* weights the source tiles by the density ratio of target to source descriptors estimated with a logistic domain classifier, clipped at its 99th percentile [21], and *CORAL* re-colours the source descriptors with the target covariance before ridge regression [23].

Each operator’s hyper-parameters (bandwidth σ , k , ridge and MLP penalties, number of trees, early-stopping epoch) are chosen on the validation city of the split by the root-mean-square standardised error over both parameters; the chosen values are listed in Table B.16. Stochastic operators are run with five seeds.

Behaviour of kernel transfer. Write $d_j = \|\mathbf{v}_u - \mathbf{v}_j\|^2$, $\delta_j = d_j - \min_k d_k \in [0, D_u]$, $\bar{\theta} = N^{-1} \sum_j \theta_j$ and $\sigma_\theta^2 = N^{-1} \sum_j \|\theta_j - \bar{\theta}\|^2$.

Theorem 1 (Two regimes of kernel transfer). (a) If $\epsilon_u = D_u/(2\sigma^2) < 1$, then

$$\hat{\theta}_u = \bar{\theta} - \frac{1}{2\sigma^2} \text{Cov}_j(\delta_j, \theta_j) + O(\epsilon_u^2), \quad \|\hat{\theta}_u - \bar{\theta}\| \leq \frac{D_u \sigma_\theta}{4\sigma^2} + O(\epsilon_u^2), \quad (8)$$

and $n_{\text{eff}}(u) = N(1 + O(\epsilon_u^2))$: the operator returns the source mean up to a first-order correction. (b) Let g_u be the gap between the smallest and second-smallest squared distance. Then the weight of the nearest source tile satisfies

$$\tilde{w}_{(1)} \geq (1 + (N-1)e^{-g_u/(2\sigma^2)})^{-1}, \quad (9)$$

so $\hat{\theta}_u$ tends to the label of the nearest source tile when $g_u \gg 2\sigma^2 \ln N$.

The minimum distance cancels from Eq. (7); a target far from every source tile is therefore *not* by itself pushed to either regime. Which regime applies is decided by the selected bandwidth relative to the spread D_u and the gap g_u , and we measure it (Section 5.4) rather than assume it. Both regimes carry a failure mode under shift: the flat regime returns the source mean and ignores the target descriptor; the nearest-neighbour regime copies one source tile and transmits its label noise undamped (Proposition 2 with $n_{\text{eff}} = 1$).

4.5. Few-shot correction

With k target tiles surveyed, the zero-shot prediction $g(\mathbf{v})$ is corrected either by the mean anchor residual (*bias*) or by a ridge model of the residual on the descriptors (*residual*, penalty 10 on normalised descriptors). Anchors are drawn uniformly at random, as the k tiles closest to a random tile (*contiguous*), which mimics a single drive-test area [24], or by a *design* that uses no measurement: the k tiles that greedily maximise the facility-location objective $\sum_i \max_{j \in T} \exp(-\|\mathbf{v}_i - \mathbf{v}_j\|^2/2h^2)$ over the target tiles, with h the median pairwise descriptor distance; the objective is monotone submodular, so the greedy set is within a factor $(1 - 1/e)$ of the optimum [28, 26]. Evaluation is always on the unsurveyed tiles, anchors are shared by all operators, and ten random draws are made per $k \in \{3, 5, 10, 25, 50, 100\}$.

4.6. Site-aware link transfer

Proposition 1 shows that any tile model assigns every pixel to its nearest site and describes interference through the exponent alone. As a transfer model free of this restriction we predict the path loss of each (pixel, site) link directly. The features are computed from the scene geometry only, so they are available for a city without any measurement. The building heights are rasterised at 2 m and the straight segment from the site to the pixel is sampled at 96 interior points, where the building height $H(s)$ is compared with the line-of-sight height $h_{\text{los}}(s)$. The link features are the 3-D and horizontal distances, the site height and the elevation angle; the fraction of the segment with $H > h_{\text{los}}$, the number of separate obstructions and the length over footprints; the largest excess height $\max_s(H - h_{\text{los}})$; the single knife-edge loss $J(v) = 6.9 + 20 \log_{10}(\sqrt{(v-0.1)^2 + 1} + v - 0.1)$ for $v > -0.78$ (zero otherwise) of the most obstructing point, with $v = h \sqrt{2(1/d_1 + 1/d_2)/\lambda}$ [50]; whether the receiver lies inside a footprint and that building’s height; and the built fraction and mean building height in a 100 m square window centred on the receiver and the built fraction in a 200 m window centred on the site. The 32 descriptors of the receiver’s tile are appended.

The ray tracer reports only links below the 150 dB cut-off, so a regressor trained on them never learns that a distant, obstructed site is not received at all; it would predict a finite loss for every link and overstate the reach of every site. We therefore use a hurdle model: a gradient-boosted classifier [49] of whether a (pixel, site) pair lies below the cut-off, trained on 120,000 pairs of each training city including the pairs beyond the cut-off, and a gradient-boosted regressor of the path loss, trained on 120,000 ray-traced links of each training city. A link is predicted beyond the cut-off (zero received power) when the classifier gives a probability below one half, and otherwise its path loss is the regression. Both models use early stopping on the validation city (learning rate 0.1, 127 leaves, at most 1500 trees, five seeds). The path-loss error of the link model is that of the regressor over the ray-traced links; the regressor alone, which keeps every link, is reported as an ablation.

Survey design for site calibration. In the few-shot variant the links observed in k surveyed tiles, which carry their serving site as minimisation-of-drive-tests (MDT) reports do [24, 25], calibrate a per-site offset. Model the residual of a link of site s as $o_s + \epsilon$, with independent site offsets o_s of prior mean $\bar{o}$ and variance τ_o^2 and link noise of variance σ_ϵ^2 , and let $\kappa = \sigma_\epsilon^2/\tau_o^2$. With n_s observed links of site s and mean residual $\bar{r}_s$, the posterior mean of o_s is $(n_s \bar{r}_s + \kappa \bar{o})/(n_s + \kappa)$, which we use with $\bar{o}$ the mean residual of all observed links and $\kappa = 50$ links, and its posterior variance is $\tau_o^2 \kappa/(n_s + \kappa)$.

Proposition 5 (Survey design). Let $c_{ts} \geq 0$ be the number of links of site s observable in tile t and $n_s(T) = \sum_{t \in T} c_{ts}$ for a set T of surveyed tiles. The total reduction of posterior variance of the site offsets,

$$F(T) = \tau_o^2 \sum_s \frac{n_s(T)}{n_s(T) + \kappa}, \quad (10)$$

is monotone and submodular in T , so the greedy choice of k tiles attains $F(T_{\text{greedy}}) \geq (1 - 1/e) \max_{|T|=k} F(T)$.

Before any measurement c_{ts} is unknown; the *designed* survey uses the number of links of site s in tile t that the hurdle classifier of the zero-shot link model predicts below the cut-off. It is compared with random surveys for $k \in \{10, 25, 50\}$ (ten draws).

4.7. Network-level evaluation

For the held-out city, each operator’s tile parameters and the true site positions define a predicted path-loss map for every (pixel, site) pair through Eq. (2); the ray-traced map is the truth, with pairs beyond the 150 dB cut-off treated as zero received power. With $P_{\text{tx}} = 46$ dBm, 20 MHz bandwidth and a 7 dB noise figure ($N = -94.0$ dBm) we compute the path-loss error over all ray-traced links, the serving site, coverage at $L \in \{120, 130, 140\}$ dB, the SINR of Eq. (1) and the spectral efficiency (attenuated Shannon, factor 0.6, zero below -10 dB, capped at 4.4 b/s/Hz [32]). An SINR below -20 dB, including a pixel at which no site is predicted to be received, is set to -20 dB (outage) for the truth and every predictor alike. Mobility decisions depend on where the serving cell changes: a pixel is a cell-edge pixel when its eastern or northern neighbour has a different serving site (one pixel per boundary crossing), the *edge displacement* is the mean distance between the true and the predicted cell edges (the average of the two directed mean nearest-edge distances), and the *edge-length error* is the relative error of the number of cell-edge pixels, a proxy for the handover rate along a route.

Site planning. Selecting K of the $|\mathcal{B}|$ sites to maximise the number of covered pixels is a monotone submodular maximisation, for which the greedy algorithm attains a $(1 - 1/e)$ approximation of the optimum of the function it is given [28]; site selection with propagation models is a classical planning task [27]. We run the greedy on the *predicted* coverage at 140 dB and score the selected sites on the *true* coverage. The regret is the true coverage of the greedy plan computed on the true map minus that of the plan computed on the predicted map. For dimensioning, we also count the sites that the greedy order on the predicted map needs to cover 90% of the pixels, and the true coverage those sites achieve.

Certified coverage. A planner must decide which pixels to *promise* as covered. For a transfer model and a held-out city c , let $r_p = PL_{\text{serv}}(p) - \widehat{PL}_{\text{serv}}(p)$ be the true minus the predicted serving path loss of pixel p (the smallest path loss over the sites), and q_c the $(1 - \alpha)$ quantile of $\{r_p\}$ over the pixels of c .

Proposition 6 (City-level coverage certificate). *Let the n cities be exchangeable and the training procedure symmetric in its source cities (the validation city drawn at random). With q_c computed from the leave-one-city-out model of each city c , set $\hat{q}_t = \max_{c \neq t} q_c$ for a target city t and claim pixel p as covered at level L when $\widehat{PL}_{\text{serv}}(p) + \hat{q}_t \leq L$. Then, with probability at least $(n - 1)/n$ over the draw of the target city, at most a fraction α of its pixels are claimed but not covered.*

The guarantee is at the level of cities, the unit of shift, in the spirit of conformal prediction [29, 30]; it needs no model of the residuals. We report it for $\alpha \in \{0.05, 0.10\}$ and $L \in \{130, 140\}$ dB with the claimed and the falsely claimed share of the pixels, and compare it with the naive claim $\widehat{PL}_{\text{serv}}(p) \leq L$.

4.8. Robustness analyses

Network size. Every zero-shot predictor is also scored on the network of the first ten sites selected by the siting rule (one site per 0.9 km^2 , inter-site distance about 1020 m). *Siting policy.* The models trained on the main corpus are applied unchanged to the siting replicate; the tile model is scored with labels fitted on the replicate itself and with the labels of the same tiles from the main corpus. *Tile size.* Labels, descriptors, zero-shot transfer (median, ridge, GBDT) and the network evaluation are repeated for $G_s \in \{50, 100, 200\}$ m. *Censoring.* Links beyond the cut-off are missing, which can bias a least-squares slope. We refit the labels with the cut lowered to 140 and 130 dB and, for 300 random tiles per city, compare the least-squares slope with that of a censored-Gaussian (Tobit) model [31] that also uses the (pixel, site) pairs beyond the cut. *Second band.* The tile and link pipelines are repeated at 7.5 GHz, and the 3.5 GHz link models are applied to the second band with the free-space offset $20 \log_{10}(f_2/f_1)$, their knife-edge feature evaluated at the new wavelength. *External scenes.* The Munich, Etoile and Florence scenes distributed with Sionna RT were modelled independently of our corpus, with detailed meshes and several ITU materials (brick, concrete, marble, metal, wood). We derive building footprints and heights from their meshes by vertical ray casting on a 2 m grid, place 8 rooftop sites per scene with the rule of our corpus (minimum separation 150 m, the scenes being smaller), ray trace them with the corpus settings and evaluate, without any retraining, the tile labels, the source median and GBDT fitted on all eleven cities, and the mean of the eleven leave-one-city-out link models. The San Francisco scene has terrain and would require a terrain-following measurement surface; it is not used.

4.9. Protocol and inference

Every city is the target once (leave-one-city-out); one of the remaining cities, drawn with a fixed seed, is the validation city and the others are training cities. Tiles of one city are spatially correlated, so tile-level tests would overstate significance; the unit of inference is the target city. For each comparison we report the mean per-city difference, a 95% percentile bootstrap interval over cities [51], the exact two-sided Wilcoxon signed-rank p -value [52] and the number of cities in which the first operator is better, with Holm’s correction [53] within each family of comparisons. Standard deviations across cities use $n - 1$.

5. Results

5.1. Corpus and ray-budget convergence

The corpus contains 11 cities, 105,800 buildings, 220 sites and 8,977,400 ray-traced links (Table 1); 10,428 tiles have at least 30 links and are labelled. Figure 2 shows one city. The

share of buildings whose height is imputed from the city median ranges from 2% (New York) to 50% (London).

The radio-map solver is a Monte-Carlo method: a cell that no ray reaches reports zero power, so the set of reached cells and their values depend on the ray budget. Table 2 compares budgets with a reference run of 3×10^8 rays for one site. At the 1×10^8 rays per site used for the corpus, path loss below 120 dB agrees with the reference to 0.53 dB RMS and path loss below 140 dB to 2.0 dB with a mean difference of 0.05 dB, and the number of cells reached below 140 dB is 96.1% of the reference. The complete corpus took 4.7 h on a CPU (LLVM) backend (77 s per site).

5.2. Tile labels and their precision

Table 3 summarises the labels. City medians of the tile exponent range from 2.39 (Vienna) to 3.15 (Hamburg); their standard deviation across cities, 0.27, is only 0.3 times the typical standard deviation of the tiles within a city (0.60–0.96). Most of the variation of the tile exponent is therefore within cities, not between them. The fit residual of the log-distance law inside a tile is 11.2–12.7 dB.

Without shrinkage, the OLS exponents span -4.8 to 8.8 , the ill-conditioning predicted by Proposition 3; empirical-Bayes shrinkage (median weight 0.36) narrows this to -0.90 – 6.45 . Label precision is the decisive quantity (Table 4). The standard error assuming independent residuals is 0.141; the cluster-robust error with sites as clusters is 0.456, 3.2 times larger, and the split-half estimate extrapolated to the full data is 0.686. The exponent of a tile therefore depends strongly on which sites serve it, and a label error of this size is comparable to the differences between cities.

The descriptors carry signal (Fig. 3b): the strongest single correlation with the exponent is $r = -0.45$ for footprint density, and no descriptor exceeds $|r| = 0.45$.

5.3. Zero-shot transfer of tile parameters

Table 5 and Fig. 4 report the exponent error on every held-out city. The source median attains 0.871; kernel transfer 0.813; the Gaussian process with learned length scales 0.764; the learned regressors 0.769 (ridge), 0.761 (MLP), 0.767 (neural encoder) and 0.782 (GBDT). The domain-adaptation variants do not improve on their base learners (CORAL 0.787 against ridge 0.769; GBDT-IW 0.784 against GBDT 0.782), and the standard models are worse than the median (3GPP UMa 1.003 and 1.453, COST-231 Hata 1.054). The best operator is the MLP. For the intercept at 100 m the learned operators reduce the error from 13.8 dB (median) to 9.6 dB (GBDT).

At the level of cities (Table C.17), GBDT improves on the median by -0.089 (95% CI $[-0.132, -0.043]$, $p = 0.010$, better in 9 of 11 cities) and on kernel transfer by -0.030 ($[-0.057, -0.008]$); kernel transfer does not improve significantly on the median (-0.059 , $[-0.111, +0.005]$, $p = 0.147$), whereas learning the metric of the same kernel does: GP-ARD improves on kernel transfer by -0.049 ($[-0.075, -0.025]$, better in 9 of 11 cities). With eleven cities, none of these improvements is significant after Holm’s correction over the 12 comparisons with

the median; the only significant differences are deteriorations (3GPP UMa NLoS). The hardest target is Chicago, where the best operator reaches 0.835.

5.4. Regime of kernel transfer

Table 6 places the tuned kernel operator relative to Theorem 1. The selected bandwidths are $\sigma = 0.1$ – 0.5 and the median ϵ_u ranges from 11.4 to 315; no target tile satisfies the condition of part (a), and the largest single weight has median 0.01, so the operator is not a nearest-neighbour copy either. Where the largest bandwidth is selected, however, it averages over up to 73% of the source tiles and its predictions spread over as little as 0.18 of the spread of the target labels: it approaches the source mean although part (a) does not hold. Across the targets the prediction spread is 0.18–0.75 of that of the labels. Its disadvantage is the noise-transmission term of Proposition 2 combined with an isotropic distance in which every descriptor counts equally; the Gaussian process, which learns one length scale per descriptor, removes most of the gap to the learned regressors (Section 5.3).

5.5. Few-shot correction of tile parameters

Surveying target tiles does not help the tile exponent at small k (Table 7, Fig. 5). With ten random tiles the residual correction changes the GBDT error from 0.782 to 0.786 and improves it in only 3 of 11 cities; for ridge the change is $+0.013$ ($[+0.004, +0.020]$). The anchors’ labels carry the site-dependent error of Section 5.2, so a correction fitted to ten of them adds more variance than it removes. The error falls below the zero-shot value only from $k = 25$ surveyed tiles on (GBDT 0.760), and reaches 0.743 at $k = 100$. Contiguous survey areas are worse than random tiles at every k (GBDT at $k = 10$: 0.882), because neighbouring tiles repeat the same information. The facility-location design, which spreads the survey over the descriptor space without any measurement, gives 0.778 at $k = 10$ (-0.007 against random tiles, $[-0.022, +0.010]$, better in 7 of 11 cities); from $k = 10$ on it beats the zero-shot error. With three or five tiles the design is worse than random anchors (Table 7), and at large k the two converge.

5.6. Network-level evaluation

Table 8 and Fig. 6 evaluate the predictors where a planner uses them. The first row is the tile model with the true labels (oracle), which isolates the error of the tile model itself. The oracle has a path-loss error of 11.6 dB, selects the correct serving site for 50.1% of the pixels (the nearest-site rule of Proposition 1) and misjudges the SINR by 7.6 dB on average. Transferred tile parameters reduce the path-loss error of the source median from 14.5 dB to 12.8 dB (GBDT; -1.716 dB, better in 11 of 11 cities) but remain above the oracle, and a plan of $K = 5$ sites made on their map loses 20.6% of the pixels against 9.3% for the oracle.

Site-aware link transfer removes the ceiling of the tile model. Its path-loss error is 9.3 dB (8.8 to 9.6 dB across cities), 3.5 dB below tile transfer with GBDT (95% CI $[-3.896, -3.118]$, better in 11 of 11 cities) and 2.3 dB below the tile model with

Table 1: Corpus. Height source: share of buildings with an explicit height attribute, with a floor count, or imputed from the city median. Links: ray-traced (pixel, site) pairs below the 150 dB cut-off. Tiles: labelled tiles (≥ 30 links).

City	Buildings	Height source (%)			Median (m)	Links	Tiles
		tag	storeys	imputed			
Amsterdam	21,135	83	2	15	15.5	736,255	948
Barcelona	12,337	12	64	23	19.2	609,050	942
Berlin	6,036	15	63	22	16.0	1,009,619	957
Chicago	2,952	70	4	25	15.9	1,069,552	961
Hamburg	4,335	17	42	41	16.0	1,152,125	961
London	11,891	8	42	50	12.8	931,288	940
Munich	8,616	9	49	42	12.8	794,056	950
New York	7,126	98	1	2	18.6	827,510	960
Paris	14,383	3	64	33	19.2	529,649	910
Prague	9,907	1	70	29	16.0	783,805	960
Vienna	7,082	26	60	14	16.0	534,491	939
Total	105,800					8,977,400	10,428

Table 2: Ray-budget convergence for one site (Amsterdam, 10 m cells). For each path-loss level: number of cells reached below that level, relative to the 3×10^8 -ray reference (%; Monte-Carlo noise can exceed 100%), and RMS path-loss difference to the reference over the cells reached by both (dB). s: solver time.

Rays	s	≤ 120 dB		≤ 130 dB		≤ 140 dB		≤ 150 dB	
		cells	RMS	cells	RMS	cells	RMS	cells	RMS
1×10^7	9	95.9	1.69	82.4	2.92	75.1	4.48	74.7	6.68
3×10^7	27	99.8	0.98	92.7	1.88	87.7	3.11	86.1	4.54
1×10^8	98	100.3	0.53	98.4	1.14	96.1	2.00	94.9	3.13
3×10^8	245	100.0	0.00	100.0	0.00	100.0	0.00	100.0	0.00

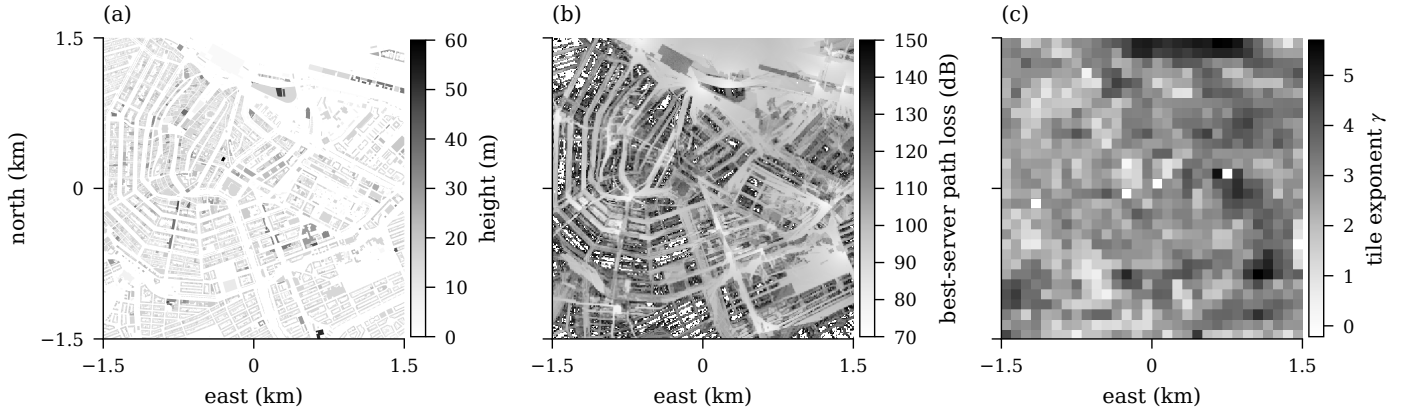


Figure 2: Amsterdam. (a) Building heights of the scene. (b) Ray-traced path loss to the best site. (c) Tile exponents $\hat{\gamma}_t$.

the *true* labels ($[-2.610, -2.075]$, 11 of 11). It selects the correct serving site for 68.8% of the pixels (+18.7 percentage points against GBDT, in 11 of 11 cities), misjudges the SINR by 5.4 dB on average and loses 0.5% of the pixels when planning $K = 5$ sites, against 20.6% for GBDT tile transfer and 9.3% for the true tile labels. The hurdle is what makes the plans reliable: the regressor alone, which keeps every link, has the same path-loss error but a planning regret of 5.6%, because it lets distant, obstructed sites cover pixels they do not reach.

Cell edges and dimensioning. Table 9 turns the maps into mobility and dimensioning quantities. Every tile model draws the

nearest-site cell pattern of Proposition 1: its cell edges lie 80 m from the true edges on average (76 m with the true labels), and their total length is in error by -93% : the ray-traced serving pattern is fragmented by buildings and a route crosses many more cell edges than the nearest-site pattern suggests. Link transfer places the edges 7 m from the truth with an edge-length error of -5% . For dimensioning, the truth needs on average 6.7 of the 20 sites to cover 90% of the pixels at 140 dB. The tile models, which cannot represent the obstruction of distant sites, predict that a single site suffices (GBDT; 1.1 sites with the true labels, averages over the cities), and the sites they choose cover only 47.4% of the pixels in truth. Link transfer chooses 3.8

Table 3: Tile labels per city. City-wide fit: OLS over all links. Tile labels: median and standard deviation of $\hat{\gamma}_i$ and median $\hat{P}_{L_0,i}$ (dB, at $d_0 = 100$ m); $\hat{\tau}$: between-tile standard deviation of the slope; w_i : shrinkage weight; RMSE: link-weighted fit residual (dB).

City	City-wide fit		Tile labels			$\hat{\tau}$	med. w_i	RMSE
	γ	P_{L_0}	med. γ	s.d. γ	med. P_{L_0}			
Amsterdam	2.47	100.5	2.84	0.93	100.9	1.06	0.27	11.3
Barcelona	1.92	105.4	2.43	0.96	105.5	1.14	0.28	11.6
Berlin	2.65	98.2	2.82	0.81	97.5	1.00	0.36	11.6
Chicago	0.81	115.6	2.79	0.83	97.2	1.19	0.54	12.7
Hamburg	2.14	100.2	3.15	0.72	92.2	0.94	0.37	11.4
London	1.72	104.4	2.40	0.60	103.3	0.79	0.42	11.2
Munich	2.12	104.0	3.00	0.80	99.8	0.92	0.33	11.2
New York	1.11	115.9	2.69	0.79	102.2	1.18	0.53	12.6
Paris	2.06	103.6	2.41	0.95	106.7	1.13	0.29	11.8
Prague	2.48	100.9	2.92	0.77	99.0	0.95	0.37	11.4
Vienna	2.29	106.1	2.39	0.92	108.4	1.09	0.32	11.4

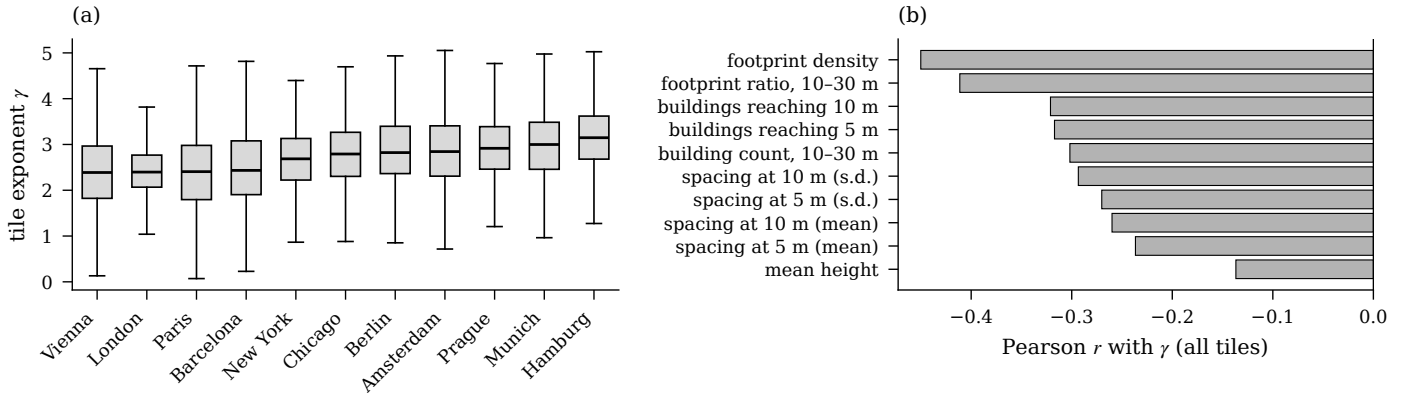


Figure 3: (a) Tile exponents per city (box: quartiles; whiskers: 1.5 times the interquartile range). (b) The ten descriptors most correlated with the tile exponent over all 10,428 tiles.

Table 4: Standard error of the tile labels: independent residuals (iid), cluster-robust over sites (cluster), and split-half over random halves of the sites extrapolated to all sites (split). Root mean square over tiles.

City	sites/tile	se(γ)			se(P_{L_0}) (dB)		
		iid	cluster	split	iid	cluster	split
Amsterdam	18	0.140	0.440	0.600	1.42	5.06	6.90
Barcelona	18	0.178	0.470	0.592	1.73	5.31	6.91
Berlin	19	0.131	0.445	0.583	1.50	5.34	6.99
Chicago	19	0.117	0.545	1.006	1.33	6.73	12.38
Hamburg	20	0.098	0.402	0.629	1.09	4.77	7.45
London	19	0.105	0.353	0.537	1.18	4.39	6.62
Munich	18	0.131	0.397	0.559	1.41	4.76	6.75
New York	17	0.142	0.541	0.890	1.54	6.65	10.40
Paris	17	0.187	0.488	0.651	1.91	5.57	7.56
Prague	19	0.115	0.416	0.667	1.25	4.84	7.45
Vienna	16	0.179	0.481	0.674	1.80	5.42	7.69
RMS		0.141	0.456	0.686	1.49	5.40	8.10

sites on average, which cover 81.6%; it still under-dimensions the network, but by far less (the regressor alone: a single site, 56.9%).

The model relies on the propagation features, which account for 97% of the gain of its trees (Table 10); the single most important feature is the knife-edge loss (23%), and the three most important together account for 60%.

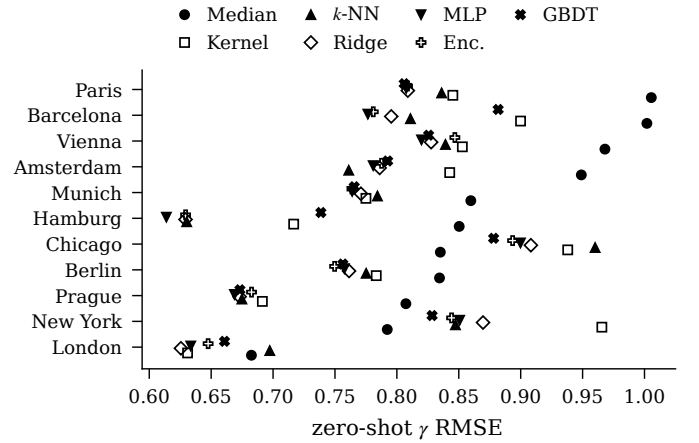


Figure 4: Zero-shot exponent RMSE per held-out city, ordered by the error of the source median.

5.7. Survey design for site calibration

Calibrating the per-site offsets of the link model from surveyed tiles does not improve it (Table 11). With ten random tiles the path-loss error rises from 9.33 dB to 9.46 dB (+0.134 dB, worse in every city), and with fifty designed tiles it is 9.44 dB; the design of Proposition 5, which maximises the vari-

Table 5: Zero-shot exponent RMSE per held-out city (mean of five seeds for stochastic operators; best per row in bold). Last row: mean P_{L_0} RMSE (dB).

Target city	Median	UMa-L	UMa-N	COST	Kernel	GP	k -NN	Ridge	CORAL	MLP	Enc.	GBDT	IW
Amsterdam	0.949	1.146	1.394	1.035	0.843	0.784	0.761	0.786	0.820	0.781	0.788	0.793	0.801
Barcelona	1.002	0.998	1.730	1.318	0.900	0.777	0.811	0.795	0.817	0.777	0.781	0.882	0.895
Berlin	0.834	1.052	1.319	0.946	0.783	0.751	0.775	0.761	0.877	0.758	0.750	0.756	0.756
Chicago	0.835	1.006	1.414	1.031	0.938	0.894	0.960	0.908	0.876	0.900	0.893	0.878	0.871
Hamburg	0.850	1.162	1.078	0.770	0.717	0.631	0.630	0.629	0.616	0.614	0.629	0.739	0.668
London	0.683	0.639	1.599	1.097	0.631	0.637	0.697	0.626	0.624	0.634	0.648	0.661	0.669
Munich	0.860	1.120	1.229	0.875	0.775	0.777	0.784	0.771	0.806	0.764	0.764	0.766	0.766
New York	0.792	0.907	1.489	1.064	0.966	0.842	0.847	0.870	0.860	0.851	0.844	0.828	0.877
Paris	1.005	0.977	1.748	1.308	0.845	0.816	0.836	0.809	0.818	0.808	0.809	0.806	0.811
Prague	0.807	1.086	1.219	0.851	0.691	0.672	0.675	0.673	0.733	0.669	0.683	0.673	0.683
Vienna	0.968	0.938	1.761	1.302	0.853	0.826	0.839	0.828	0.805	0.820	0.847	0.826	0.824
Mean	0.871	1.003	1.453	1.054	0.813	0.764	0.783	0.769	0.787	0.761	0.767	0.782	0.784
S.d.	0.100	0.147	0.234	0.191	0.105	0.085	0.092	0.092	0.091	0.089	0.084	0.073	0.083
P_{L_0} (dB)	13.8	21.6	13.8	17.9	10.5	9.5	9.8	9.7	10.3	9.5	9.6	9.6	9.6

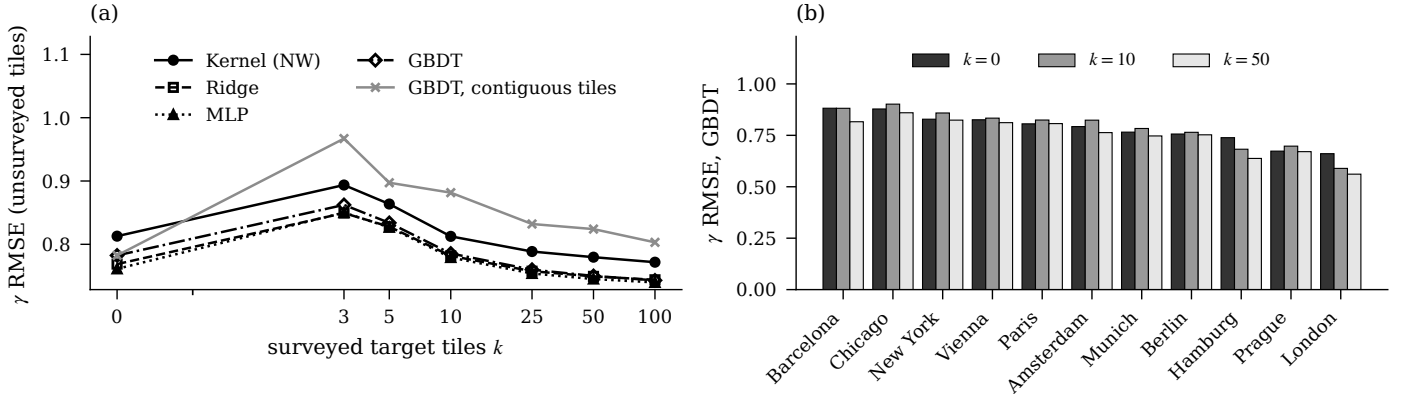


Figure 5: (a) Exponent RMSE against the number of surveyed target tiles. (b) GBDT per held-out city at $k \in \{0, 10, 50\}$.

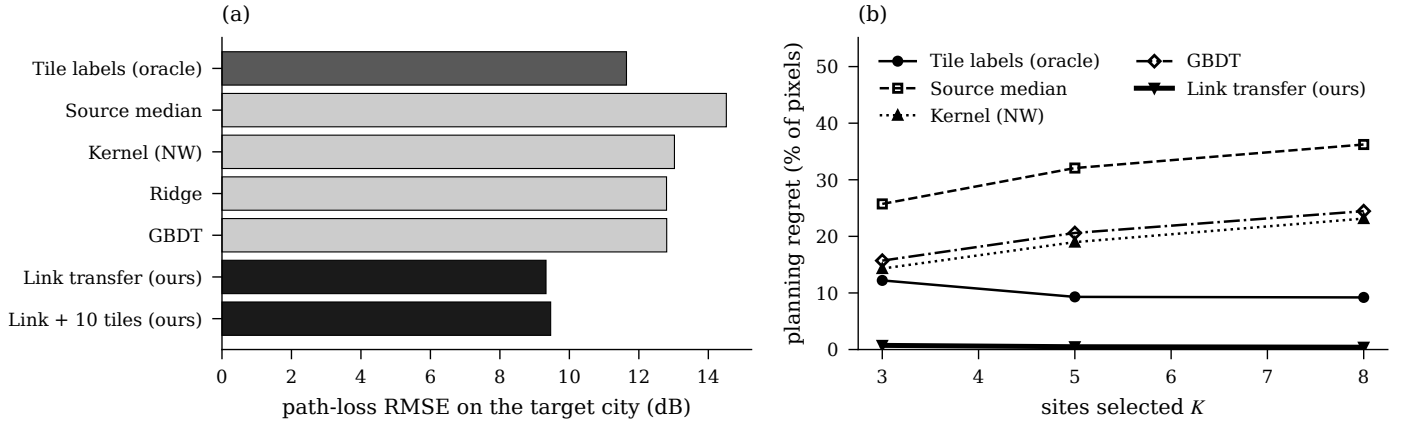


Figure 6: (a) Path-loss RMSE on the held-out cities. (b) Planning regret against the number of selected sites.

ance reduction of the offsets, is not better than random tiles. This indicates that the zero-shot link model has little systematic per-site error left for a survey of this size to remove, while an offset estimated from the links of a few tiles transmits their local residuals to the whole cell. A survey is better spent on the target quantity itself, as the next section does.

5.8. Certified coverage

Table 12 applies Proposition 6. For every predictor the bound $\hat{q}_t$ holds in 10 of 11 cities, as the guarantee of at least 10/11 requires. At $\alpha = 0.1$ and 140 dB, the certificate of link transfer needs a margin of only 9.8 dB and certifies 92.0% of the pixels, with 1.79% falsely claimed; the true covered share is

Table 6: Kernel transfer on each held-out city: selected bandwidth; median $\epsilon_u = D_u/2\sigma^2$ and share of tiles with $\epsilon_u < 1$ (Theorem 1a); median effective-neighbourhood fraction; median largest weight (Theorem 1b); standard deviation of the predictions over that of the labels.

Target	σ	med. ϵ_u	$\epsilon_u < 1$ (%)	n_{eff}/N (%)	med. w_{max}	spread
Amsterdam	0.1	315	0	2.19	0.020	0.49
Barcelona	0.5	11.4	0	70.90	0.000	0.18
Berlin	0.1	294	0	5.07	0.009	0.44
Chicago	0.1	307	0	0.45	0.073	0.65
Hamburg	0.5	12.4	0	73.05	0.000	0.27
London	0.2	75.8	0	22.15	0.001	0.60
Munich	0.2	75.3	0	27.00	0.001	0.38
New York	0.1	277	0	0.04	0.467	0.75
Paris	0.1	304	0	5.00	0.008	0.39
Prague	0.1	299	0	6.28	0.007	0.43
Vienna	0.1	294	0	5.54	0.008	0.43

Table 7: Few-shot correction: exponent RMSE on the unsurveyed tiles, mean over cities (and ten anchor draws for random and contiguous anchors). Left: residual correction with random anchors. Right: GBDT with a bias correction, with contiguous anchors and with the facility-location design.

k	residual correction, random tiles				GBDT		
	Kernel	Ridge	MLP	GBDT	bias	contig.	design
0	0.813	0.769	0.761	0.782	0.782	0.782	0.782
3	0.894	0.849	0.850	0.862	0.864	0.967	0.905
5	0.864	0.828	0.827	0.834	0.836	0.897	0.840
10	0.813	0.782	0.779	0.786	0.790	0.882	0.778
25	0.789	0.758	0.754	0.760	0.770	0.832	0.759
50	0.780	0.749	0.745	0.750	0.765	0.824	0.755
100	0.772	0.744	0.740	0.743	0.762	0.803	0.749

96.3%. Tile transfer needs larger margins and certifies less: GBDT 78.3% (margin 17.6 dB), the source median 60.5% and even the true tile labels 88.2%. The naive claim without a margin covers almost every pixel but is wrong for 3.11% (link) to 3.68% (GBDT) of them, with no stated risk. The accuracy of a transfer model thus translates directly into the coverage an operator can promise at a stated risk.

5.9. Robustness

Tile size. Table 13 repeats the tile pipeline for $G_s \in \{50, 100, 200\}$ m; $G_s = 100$ m reproduces the main results exactly. Larger tiles give more precise labels (cluster-robust standard error 0.536, 0.453 and 0.366) and a smaller exponent error (GBDT 0.888, 0.782 and 0.619), but a coarser map: the path-loss error of the oracle grows from 11.2 dB to 12.1 dB, and that of GBDT tile transfer stays between 12.7 dB and 13.1 dB. The serving-site accuracy of the tile model is 50.1–50.1% at every size, as Proposition 1 implies. No tile size approaches link transfer.

Path-loss cut-off. Between 35.5% and 66.4% of the (pixel, site) pairs of the labelled tiles lie beyond the cut-off (Table 14). The labels depend on this dynamic range: lowering the cut-off to 140 dB changes the median tile exponent by -0.583 and to 130 dB by -1.119 , and a Tobit fit that also uses the censored

pairs gives slopes larger than the least-squares slopes by a median of 2.11 to 4.49 across the cities (3,300 tiles). The tile exponent thus depends on the dynamic range of the measurements, not only on the environment, which is a further reason to model links, and to model the cut-off explicitly as the hurdle of the link model does.

Network size. On the network of the first 10 sites (Table 15), the ranking is unchanged: the oracle has a path-loss error of 11.7 dB, GBDT tile transfer 13.0 dB and link transfer 9.2 dB, with serving-site accuracies of 51.3%, 51.3% and 68.3%.

5.10. Statistical summary

Tables C.17 and C.18 (Appendix C) list all 56 city-level comparisons. Link transfer improves on GBDT tile transfer in path loss, serving site, coverage agreement and planning regret in all 11 cities, and on the tile model with the true labels in path loss, serving site and planning regret in all 11 cities; these comparisons remain significant after Holm’s correction. At the parameter level the evidence is weaker: of the 12 comparisons of a transfer operator with the source median, no improvement survives Holm’s correction at eleven cities, whereas the gain of GP-ARD over kernel transfer does.

6. Discussion

The tile exponent is not an environmental constant. The central premise of descriptor-based tile transfer is that a tile has a path-loss exponent determined by its environment, which can be read from descriptors and carried to similar tiles elsewhere. On this corpus the premise is weak, for three measured reasons. Two estimates of the same tile’s exponent from disjoint halves of the sites differ by 1.37 in standard deviation, so the exponent depends on which sites serve the tile; it depends on the dynamic range of the measurements, shifting by -0.583 when the cut-off is lowered by 10 dB; and the cross-city spread of the city medians (0.27) is small against the spread within a city. A tile is seen by each site through a different street canyon, and the exponent fitted over all of them mixes the geometry of the tile with that of the paths reaching it. This explains why learned operators improve on the source median only modestly, why few-shot correction of the exponent needs tens of surveyed tiles before it helps, and why a planning tool fed with even the *true* tile labels assigns half of the pixels to the wrong site and draws cell edges 76 m away from the true ones.

What to transfer. The quantity that transfers is the effect of the geometry along a link, not a per-tile summary of it. Site-aware link transfer uses the same descriptors, the same training cities and the same protocol, but conditions on the path between site and receiver; it improves path loss, serving site, coverage and planning in every held-out city and outperforms the tile model with perfect labels. Its most important features, the knife-edge loss and the height of the obstruction above the line of sight, are the quantities of classical diffraction theory [50], so the model remains interpretable in physical terms. Two design choices matter for planning. First, the model must represent the absence

Table 8: Network-level evaluation on the held-out cities (mean over cities). Oracle: tile model with the true labels. Serving site: share of pixels with the correct serving site. Coverage: agreement of covered pixels (serving path loss ≤ 130 dB) and error of the covered share. SINR: mean absolute error and Kolmogorov–Smirnov distance of the distributions. SE: error of mean spectral efficiency (b/s/Hz). Regret: coverage lost by planning $K = 5$ sites on the predicted map instead of the true map. True covered share: 89.3%; true mean spectral efficiency: 1.47 b/s/Hz.

Predictor	Path loss (dB)		Serving site (%)	Coverage 130 dB (%)		SINR		SE	Regret $K=5$ (%)
	RMSE	bias		agree	Δ	MAE (dB)	KS	Δ	
Tile labels (oracle)	11.64	+0.00	50.1	89.6	6.0	7.56	0.304	−0.652	9.3
Source median	14.52	+3.68	50.1	87.9	8.9	7.18	0.290	−0.668	32.1
3GPP UMa NLoS	22.75	+17.29	50.1	70.2	−16.9	6.92	0.118	−0.213	8.0
COST-231 Hata	25.29	+20.89	50.1	52.8	−40.1	7.14	0.218	−0.494	9.9
Kernel (NW)	13.02	+1.30	50.1	88.2	8.4	7.47	0.296	−0.675	19.0
GP-ARD	12.80	+1.02	50.1	88.1	7.5	7.56	0.298	−0.676	17.9
k -NN	12.83	+1.12	50.1	88.4	8.5	7.56	0.291	−0.665	28.8
Ridge	12.80	+1.06	50.1	87.9	7.6	7.54	0.298	−0.674	18.9
CORAL + ridge	13.08	+2.38	50.1	87.3	5.5	7.63	0.316	−0.708	18.3
MLP	12.76	+1.02	50.1	88.2	7.8	7.57	0.297	−0.673	26.5
Neural encoder	12.79	+0.86	50.1	88.4	8.5	7.54	0.294	−0.668	25.8
GBDT	12.80	+0.97	50.1	88.3	8.2	7.44	0.294	−0.671	20.6
GBDT-IW	12.82	+0.89	50.1	88.3	8.1	7.44	0.293	−0.669	18.7
GBDT + 10 tiles	12.75	+0.96	50.1	88.5	7.7	7.56	0.299	−0.673	20.4
GBDT + 10 designed tiles	12.73	+1.25	50.1	88.3	7.3	7.52	0.290	−0.643	19.8
Link transfer (ours)	9.33	−0.01	68.8	90.4	2.4	5.37	0.048	−0.078	0.5
Link, regressor only	9.33	−0.01	68.9	90.4	2.5	5.83	0.162	−0.322	5.6
Link + 10 tiles (ours)	9.46	+0.11	65.9	90.3	2.5	5.52	0.047	−0.067	0.5
Link + 10 designed tiles (ours)	9.52	−0.15	65.6	90.1	2.6	5.65	0.051	−0.071	0.9

Table 9: Cell edges and dimensioning (mean over held-out cities). Disp.: mean distance between true and predicted cell edges; length: relative error of the number of cell-edge pixels. Plan: sites chosen greedily on the predicted map until it covers 90% of the pixels at 140 dB, and the true coverage of those sites; truth: sites the same procedure needs on the true map.

Predictor	Cell edges		Plan for 90%	
	disp. (m)	length (%)	sites	true cov. (%)
Tile labels (oracle)	76	−92	1.1	49.7
Source median	80	−93	1.0	56.4
3GPP UMa NLoS	80	−93	5.6	84.0
Kernel (NW)	80	−93	1.0	51.9
GP-ARD	80	−93	1.0	50.7
Ridge	80	−93	1.0	50.8
GBDT	80	−93	1.0	47.4
GBDT + 10 tiles	80	−93	1.0	49.3
Link transfer (ours)	7	−5	3.8	81.6
Link, regressor only	7	−4	1.0	56.9
Link + 10 designed tiles (ours)	7	−3	3.9	81.9
Truth	0	0	6.7	90+

of a link: the hurdle, which predicts whether a site is received at all, reduces the planning regret from 5.6% to 0.5% at the same path-loss error. Second, its errors must be stated at the level of the decision: the city-level certificate turns the leave-one-city-out residuals into a coverage promise with a guaranteed risk, and the more accurate the model, the more of the city it can promise (92.0% against 78.3% for tile transfer). For a digital twin this means that the transferable representation should be attached to links (or to site–tile pairs), which is also what a planning tool evaluates.

Table 10: Link transfer: the ten features with the largest share of the total split gain (mean over folds and seeds).

Feature	gain share (%)
knife-edge loss	22.9
max. excess height	19.7
buildings crossed	17.4
log distance	11.7
blocked fraction	8.5
length over buildings	5.9
horizontal distance	2.4
built fraction, 50 m	1.8
receiver building height	1.8
site height	1.6

Table 11: Per-site calibration of the link model from k surveyed tiles: random tiles (mean of ten draws) and the survey design of Proposition 5. PL: path-loss RMSE; serv.: serving-site accuracy; regret: $K = 5$. Means over the held-out cities. Zero-shot link model: 9.33 dB, 68.8%, 0.5%.

k	Random survey			Designed survey		
	PL (dB)	serv. (%)	regret (%)	PL (dB)	serv. (%)	regret (%)
10	9.46	65.9	0.5	9.52	65.6	0.9
25	9.35	67.6	0.5	9.46	66.0	0.7
50	9.30	69.1	0.5	9.44	67.0	0.5

Kernel transfer. Theorem 1 identifies two limits of kernel transfer; the tuned operator is in neither, but with the bandwidths selected on two targets it approaches the source mean without reaching the flat regime. Its disadvantage against the learned regressors is the noise-transmission term of Proposi-

Table 12: Certified coverage (Proposition 6), mean over held-out cities. $\hat{q}_t$: margin from the other cities; valid: cities in which $q_t \leq \hat{q}_t$ (of 11); $r_p \leq \hat{q}_t$: share of pixels within the margin; certified claim: share claimed covered and share claimed but not covered; naive claim: without a margin; covered: true covered share.

Predictor	$\hat{q}_t$ (dB)	valid	$r_p \leq \hat{q}_t$ (%)	Certified claim		Naive claim		covered (%)
				claimed (%)	false (%)	claimed (%)	false (%)	
$\alpha = 0.1, L = 140$ dB								
Tile labels (oracle)	13.4	10	93.9	88.2	1.79	99.9	3.62	96.3
Source median	22.2	10	95.5	60.5	1.60	100.0	3.68	96.3
3GPP UMa NLoS	14.6	10	95.3	55.6	1.32	95.5	3.48	96.3
Kernel (NW)	19.3	10	95.7	72.3	1.44	100.0	3.68	96.3
Ridge	17.8	10	95.1	77.6	1.60	100.0	3.68	96.3
GBDT	17.6	10	95.1	78.3	1.66	100.0	3.68	96.3
Link transfer (ours)	9.8	10	93.9	92.0	1.79	98.3	3.11	96.3
Link, regressor only	10.3	10	94.0	91.3	1.69	100.0	3.68	96.3
$\alpha = 0.05, L = 140$ dB								
Tile labels (oracle)	18.7	10	97.3	71.0	0.75	99.9	3.62	96.3
Source median	27.5	10	98.0	32.4	0.64	100.0	3.68	96.3
3GPP UMa NLoS	20.4	10	97.9	34.2	0.63	95.5	3.48	96.3
Kernel (NW)	24.5	10	98.1	48.8	0.50	100.0	3.68	96.3
Ridge	22.5	10	97.7	57.8	0.64	100.0	3.68	96.3
GBDT	22.3	10	97.6	58.8	0.68	100.0	3.68	96.3
Link transfer (ours)	14.5	10	97.1	82.7	0.82	98.3	3.11	96.3
Link, regressor only	14.7	10	97.1	82.3	0.78	100.0	3.68	96.3

Table 13: Tile size. Tiles per city, median shrinkage weight and standard error of the exponent (iid and cluster-robust, RMS over tiles); zero-shot exponent RMSE of the source median and GBDT; network path-loss RMSE of the tile labels (oracle) and of GBDT tile transfer; serving-site accuracy of the tile model. Means over the held-out cities.

G_s (m)	tiles/city	med. w_t	se(γ)		γ RMSE		PL RMSE (dB)		serv. (%)
			iid	cluster	median	GBDT	oracle	GBDT	
50	3470	0.38	0.234	0.536	0.985	0.888	11.19	12.67	50.1
100	948	0.37	0.138	0.453	0.871	0.782	11.64	12.80	50.1
200	244	0.38	0.086	0.366	0.716	0.619	12.09	13.05	50.1

Table 14: Path-loss cut-off. Share of (pixel, site) pairs of the labelled tiles beyond the 150 dB cut-off and of links above 140 dB; median change of the tile exponent when the cut-off is lowered to 140 and 130 dB; for random tiles, median and 90th percentile of the Tobit minus the least-squares slope.

City	censored	Lower cut-off			Tobit vs least squares		
	pairs (%)	links >140 (%)	$\Delta\gamma_{140}$	$\Delta\gamma_{130}$	tiles	med. $\Delta\gamma$	q90
Amsterdam	56.8	18.4	-0.572	-1.141	300	+3.746	+6.802
Barcelona	61.0	18.7	-0.486	-0.893	300	+4.284	+7.509
Berlin	43.5	17.3	-0.646	-1.261	300	+3.256	+6.938
Chicago	40.5	17.9	-0.640	-1.226	300	+3.536	+6.698
Hamburg	35.5	15.0	-0.504	-1.066	300	+2.113	+3.886
London	46.9	14.6	-0.470	-0.855	300	+2.946	+5.014
Munich	54.7	20.1	-0.583	-1.184	300	+3.157	+5.333
New York	53.8	23.8	-0.621	-1.119	300	+4.492	+7.343
Paris	63.1	18.3	-0.598	-1.006	300	+4.266	+8.338
Prague	55.8	21.5	-0.729	-1.366	300	+3.500	+5.315
Vienna	66.4	22.9	-0.527	-0.980	300	+4.146	+7.915

tion 2 combined with an isotropic distance: learning one length scale per descriptor (GP-ARD) removes most of the gap, which locates the weakness in the metric rather than in kernel averaging as such.

Implications for planning. Four practical conclusions follow. First, a transfer model for a digital twin should be validated

at the network level; parameter errors of similar size lead to very different planning losses (Table 8), and a uniformly biased model such as the 3GPP UMa NLoS constants has the largest path-loss error of all (22.8 dB) yet a planning regret of 8.0%, below that of transferred tile parameters, because a uniform bias does not change which sites cover the most pixels. Sec-

Table 15: Network size: the same predictors on the 20-site and on the 10-site network (mean over held-out cities). PL: path-loss RMSE; serv.: serving-site accuracy; SINR: mean absolute error; regret: $K = 5$.

Predictor	20 sites				10 sites			
	PL (dB)	serv. (%)	SINR (dB)	regret (%)	PL (dB)	serv. (%)	SINR (dB)	regret (%)
Tile labels (oracle)	11.64	50.1	7.56	9.3	11.67	51.3	8.05	6.3
Source median	14.52	50.1	7.18	32.1	14.91	51.3	7.85	27.0
3GPP UMa NLoS	22.75	50.1	6.92	8.0	23.87	51.3	8.04	4.5
Kernel (NW)	13.02	50.1	7.47	19.0	13.20	51.3	8.07	18.7
GP-ARD	12.80	50.1	7.56	17.9	12.95	51.3	8.12	12.7
Ridge	12.80	50.1	7.54	18.9	12.96	51.3	8.11	14.7
GBDT	12.80	50.1	7.44	20.6	12.95	51.3	8.04	16.1
Link transfer (ours)	9.33	68.8	5.37	0.5	9.22	68.3	6.22	0.4

ond, models trained on links below a receiver cut-off overstate the reach of distant sites unless the cut-off is modelled; every zero-shot tile model with transferred parameters predicts that a single site covers 90% of a city. Third, small surveys are not an efficient way to calibrate either the tile exponent or per-site offsets; the survey is better used to certify coverage. Fourth, coverage claims should come with a margin estimated across cities: without it, 3.68% of the pixels promised by tile transfer are not covered.

Limitations. The path loss is ray traced rather than measured, so absolute values depend on the solver, the ITU material models and the level-of-detail-1 geometry; the results are comparisons under identical conditions. The external scenes test a different geometry source and different materials but the same solver, and the siting replicate and the second band test other sites and another frequency; none replaces validation against measurements, which the public drive-test data we could access do not allow because they lack site positions. Between 2% and 50% of the building heights are imputed per city. The corpus has flat terrain, 20 rooftop sites per city with isotropic antennas, full load and a fixed link budget. Tile labels come from the same ray tracer as the network truth, so the oracle is the reference for tile models with exact labels on this corpus, not a statement about measured data.

7. Conclusion

We built reproducible ray-traced corpora of 11 cities in two bands for cross-city transfer of path-loss models and evaluated transfer where a network planner uses it. Tile labels of the log-distance law are ill-conditioned and, once their correlated errors are accounted for, imprecise: the cluster-robust standard error of the exponent is 3.2 times the naive one, and the exponent depends on the serving sites, the tile size and the cut-off of the data. Transferring tile parameters with learned regressors improves on the source median; kernel transfer does not improve on it significantly, and our analysis of its limiting regimes and the gain from learning its metric locate its weakness in the isotropic distance. At the network level, even perfect tile parameters leave a path-loss error of 11.6 dB, assign only 50.1% of the pixels to the correct site and misplace cell

edges by 76 m. Site-aware hurdle link transfer, which predicts each link and whether it exists from its obstruction geometry, reduces the path-loss error to 9.3 dB, raises serving-site accuracy to 68.8% and lowers the planning regret in every held-out city, and a city-level certificate turns its accuracy into coverage that can be promised at a stated risk.

Future work should validate these findings on measured data with known site positions, extend the corpora to terrain, antenna patterns and further bands, and study certificates that adapt to the target city from a small survey.

CRedit authorship contribution statement

Abraheem Rashid: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft. **Faisal Iradat:** Software, Validation, Investigation. **Waseem Iqbal:** Supervision, Writing – review & editing. **Yawar Abbas Bangash:** Formal analysis, Writing – review & editing. **Bilgin Metin:** Supervision, Writing – review & editing.

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with software development, data analysis scripting and the drafting and editing of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The code, the scene definitions, the ray-traced path-loss corpus, the tile tables and every result file are available at <https://github.com/Abraheem13/terra-net>. Running `make reproduce` regenerates all numbers, tables and figures of this article and the article itself. Building footprints are from Overture Maps (release 2026-09-23.0), © OpenStreetMap contributors, Overture Maps Foundation, under the Open Database License.

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Appendix A. Proofs

Proposition 1. With the parameters of tile t for every site, $P_{pb} = P_{tx} - P_{L_{0,t}} - 10\gamma_t \log_{10}(d_{pb}/d_0)$ is strictly decreasing in d_{pb} when $\gamma_t > 0$, which gives (i). Writing $10^{P_{pb}/10} = C(d_{pb}/d_0)^{-\gamma_t}$ with $C = 10^{(P_{tx} - P_{L_{0,t}})/10}$ in Eq. (1) and dividing numerator and denominator by C gives Eq. (3). $\square$

Proposition 2. With $\hat{\theta}(\mathbf{v}) = \sum_j w_j(\mathbf{v})(\theta^*(\mathbf{v}_j) + \eta_j)$ and $\sum_j w_j = 1$, $\hat{\theta} - \theta^*(\mathbf{v}) = [\sum_j w_j \theta^*(\mathbf{v}_j) - \theta^*(\mathbf{v})] + \sum_j w_j \eta_j$. The weights depend on descriptors only and $\mathbb{E}[\eta_j] = \mathbf{0}$ independently of the descriptors, so the cross term has zero expectation, and independence across j gives $\mathbb{E}[\|\sum_j w_j \eta_j\|^2] = \text{tr}(\Sigma_\eta) \sum_j w_j^2$. $\square$

Proposition 3. $\mathbf{A}^\top \mathbf{A} = [\sum_{x_i} x_i^2 \sum_n x_i]$ has determinant $n \sum x_i^2 - (\sum x_i)^2 = n^2 \text{Var}(x)$, and the (1, 1) entry of its inverse is $n/(n^2 \text{Var}(x))$, so $\text{Var}(\hat{\gamma}_{\text{OLS}}) = \sigma_\epsilon^2/(n \text{Var}(x))$. $\square$

Proposition 4. Given the true slope, b_t has mean γ_t and variance v_t ; the true slopes have mean μ and variance τ^2 . For fixed w , $\hat{\gamma}_t - \gamma_t = (1 - w)(b_t - \gamma_t) + w(\mu - \gamma_t)$; the two terms are uncorrelated, so the mean squared error is $(1 - w)^2 v_t + w^2 \tau^2$, a convex quadratic in w minimised at $w_t = v_t/(v_t + \tau^2)$ with value $v_t \tau^2/(v_t + \tau^2)$, which is at most $\min(v_t, \tau^2)$. With centred x , the ridge problem $\min_\gamma \sum_i (y_i - \bar{y} - \gamma x_i)^2 + \lambda(\gamma - \mu)^2$ has solution $(S_{xy} + \lambda\mu)/(S_{xx} + \lambda) = (1 - w)b_t + w\mu$ with $w = \lambda/(\lambda + S_{xx})$; setting $\lambda_t = S_{xx} v_t/\tau^2$ gives $w = v_t/(v_t + \tau^2) = w_t$. $\square$

Theorem 1, part (a). Put $\alpha = 1/(2\sigma^2)$. The factor $e^{-\alpha \min_k d_k}$ cancels in Eq. (7), so $\tilde{w}_j = e^{-\alpha \delta_j} / \sum_k e^{-\alpha \delta_k}$ with $\alpha \delta_j \leq \epsilon_u < 1$. By Taylor’s theorem, $e^{-\alpha \delta_j} = 1 - \alpha \delta_j + r_j$ with $0 \leq r_j \leq \epsilon_u^2/2$, hence $\sum_k e^{-\alpha \delta_k} = N(1 - \alpha \bar{\delta}) + O(N\epsilon_u^2)$ and

$$\tilde{w}_j = \frac{1}{N}(1 - \alpha(\delta_j - \bar{\delta})) + O(\epsilon_u^2/N).$$

Then $\hat{\theta}_u = \sum_j \tilde{w}_j \theta_j = \bar{\theta} - \alpha \text{Cov}_j(\delta_j, \theta_j) + O(\epsilon_u^2)$, where the remainder is bounded by a constant times $\epsilon_u^2 \max_j \|\theta_j - \bar{\theta}\|$. By

Cauchy–Schwarz, $\|\text{Cov}_j(\delta_j, \theta_j)\| \leq \sigma_\delta \sigma_\theta$, and by Popoviciu’s inequality $\sigma_\delta \leq D_u/2$, which gives the bound in Eq. (8). Squaring the weight expansion, $\sum_j \tilde{w}_j^2 = N^{-1}(1 + \alpha^2 \sigma_\delta^2) + O(\epsilon_u^2/N)$, the linear term vanishing because $\sum_j (\delta_j - \bar{\delta}) = 0$; since $\alpha \sigma_\delta \leq \epsilon_u/2$, $n_{\text{eff}} = N(1 + O(\epsilon_u^2))$. $\square$

Theorem 1, part (b). Order the source tiles by distance, so $\delta_{(1)} = 0$ and $\delta_{(j)} \geq g_u$ for $j \geq 2$. Then $\tilde{w}_{(1)} = (1 + \sum_{j \geq 2} e^{-\alpha \delta_{(j)}})^{-1} \geq (1 + (N - 1)e^{-\alpha g_u})^{-1}$, and $\|\hat{\theta}_u - \theta_{(1)}\| = \|\sum_{j \geq 2} \tilde{w}_{(j)}(\theta_{(j)} - \theta_{(1)})\| \leq (1 - \tilde{w}_{(1)}) \max_j \|\theta_j - \theta_{(1)}\| \leq (N - 1)e^{-\alpha g_u} \max_j \|\theta_j - \theta_{(1)}\|$. $\square$

Proposition 5. $n_s(T)$ is a non-negative modular function of T and $\phi(n) = n/(n + \kappa)$ is non-decreasing and concave on $[0, \infty)$. For $A \subseteq B$ and a tile $t \notin B$, $n_s(A) \leq n_s(B)$ and concavity give $\phi(n_s(A) + c_{ts}) - \phi(n_s(A)) \geq \phi(n_s(B) + c_{ts}) - \phi(n_s(B)) \geq 0$. Summing over s shows that F is monotone and has diminishing returns, i.e. is submodular, and $F(\emptyset) = 0$; the greedy bound is the theorem of [28]. The posterior variance follows from the conjugate normal model: the precision of o_s is $1/\tau_o^2 + n_s/\sigma_\epsilon^2$, so its variance is $\tau_o^2 \kappa/(n_s + \kappa)$ and the reduction is $\tau_o^2 n_s/(n_s + \kappa)$. $\square$

Proposition 6. Each q_c is a function of city c and of the multi-set of the other cities, through the same symmetric procedure, so exchangeable cities give an exchangeable vector $(q_1, \dots, q_n)$. The event $q_t > \max_{c \neq t} q_c$ means that q_t is the strict maximum of the vector, which by exchangeability has probability at most $1/n$; hence $\Pr(q_t \leq \hat{q}_t) \geq (n - 1)/n$. On this event, a claimed pixel with $r_p \leq \hat{q}_t$ satisfies $PL_{\text{serv}}(p) = \bar{P}L_{\text{serv}}(p) + r_p \leq \bar{P}L_{\text{serv}}(p) + \hat{q}_t \leq L$, so it is covered. Claimed but uncovered pixels therefore have $r_p > \hat{q}_t \geq q_t$, and by the definition of the quantile at most a fraction α of the pixels do. $\square$

Appendix B. Selected hyper-parameters

Table B.16: Hyper-parameters selected on the validation city of each split. N : training tiles; σ : kernel bandwidth; k : neighbours; trees: median number of boosting rounds for γ over the five seeds.

Target	Validation	N	σ	k	α_{ridge}	α_{MLP}	trees
Amsterdam	Prague	8,520	0.1	50	0.01	0.01	105
Barcelona	New York	8,526	0.5	1000	100	1	9
Berlin	Munich	8,521	0.1	200	0.01	0.01	80
Chicago	Berlin	8,510	0.1	200	0.01	0.01	48
Hamburg	Chicago	8,506	0.5	1000	100	1	8
London	Amsterdam	8,540	0.2	50	0.01	0.01	102
Munich	Amsterdam	8,530	0.2	50	0.01	0.01	118
New York	Amsterdam	8,520	0.1	50	0.01	0.0001	96
Paris	Barcelona	8,576	0.1	20	0.01	0.01	97
Prague	Paris	8,558	0.1	200	0.01	0.01	85
Vienna	Munich	8,539	0.1	200	0.01	0.01	57

Appendix C. Complete city-level comparisons

Table C.17: City-level comparisons of the tile parameters ($n = 11$ held-out cities). Δ : mean difference first minus second; CI: 95% percentile bootstrap over cities; p : exact two-sided Wilcoxon signed-rank; better: cities in which the first is better; Holm: significant at 5% after Holm’s correction within the family.

Comparison	Metric	Δ	95% CI	p	better	Holm
<i>Zero-shot vs source median</i>						
CORAL + ridge vs Source median	γ RMSE	-0.085	[-0.143, -0.026]	0.024	8/11	no
COST-231 Hata vs Source median	γ RMSE	+0.183	[+0.093, +0.269]	0.005	1/11	no
Neural encoder vs Source median	γ RMSE	-0.105	[-0.158, -0.047]	0.010	9/11	no
GBDT vs Source median	γ RMSE	-0.089	[-0.132, -0.043]	0.010	9/11	no
GBDT-IW vs Source median	γ RMSE	-0.088	[-0.135, -0.034]	0.014	9/11	no
GP-ARD vs Source median	γ RMSE	-0.107	[-0.160, -0.049]	0.010	9/11	no
Kernel (NW) vs Source median	γ RMSE	-0.059	[-0.111, +0.005]	0.147	9/11	no
k -NN vs Source median	γ RMSE	-0.088	[-0.148, -0.022]	0.024	8/11	no
MLP vs Source median	γ RMSE	-0.110	[-0.166, -0.049]	0.010	9/11	no
Ridge vs Source median	γ RMSE	-0.103	[-0.158, -0.042]	0.019	9/11	no
3GPP UMa LoS vs Source median	γ RMSE	+0.131	[+0.054, +0.205]	0.042	4/11	no
3GPP UMa NLoS vs Source median	γ RMSE	+0.581	[+0.462, +0.698]	<0.001	0/11	yes
<i>Zero-shot pairs</i>						
GBDT vs Kernel (NW)	γ RMSE	-0.030	[-0.057, -0.008]	0.054	9/11	no
GBDT vs Ridge	γ RMSE	+0.014	[-0.010, +0.042]	0.831	6/11	no
Kernel (NW) vs k -NN	γ RMSE	+0.030	[-0.002, +0.062]	0.147	3/11	no
GP-ARD vs Kernel (NW)	γ RMSE	-0.049	[-0.075, -0.025]	0.005	9/11	yes
GBDT-IW vs GBDT	γ RMSE	+0.001	[-0.016, +0.016]	0.278	4/11	no
CORAL + ridge vs Ridge	γ RMSE	+0.018	[-0.004, +0.043]	0.320	5/11	no
<i>Few-shot correction</i>						
GBDT, $k=10$ vs GBDT, $k=0$	γ RMSE	+0.003	[-0.018, +0.021]	0.365	3/11	no
GBDT, $k=10$ designed vs GBDT, $k=10$	γ RMSE	-0.007	[-0.022, +0.010]	0.240	7/11	no
Kernel (NW), $k=10$ vs Kernel (NW), $k=0$	γ RMSE	+0.000	[-0.012, +0.011]	1.000	5/11	no
Kernel (NW), $k=10$ designed vs Kernel (NW), $k=10$	γ RMSE	-0.002	[-0.024, +0.028]	0.413	7/11	no
MLP, $k=10$ vs MLP, $k=0$	γ RMSE	+0.018	[+0.009, +0.026]	0.010	1/11	no
MLP, $k=10$ designed vs MLP, $k=10$	γ RMSE	-0.008	[-0.028, +0.018]	0.365	8/11	no
Ridge, $k=10$ vs Ridge, $k=0$	γ RMSE	+0.013	[+0.004, +0.020]	0.024	1/11	no
Ridge, $k=10$ designed vs Ridge, $k=10$	γ RMSE	-0.010	[-0.027, +0.011]	0.278	8/11	no

Table C.18: City-level comparisons at the network level, as in Table C.17 (Δ in percentage points for shares: serving site, coverage agreement, regret).

Comparison	Metric	Δ	95% CI	p	better	Holm
<i>Network level</i>						
GBDT vs Source median	path-loss RMSE (dB)	-1.716	[-2.264, -1.210]	<0.001	11/11	yes
Link transfer (ours) vs GBDT	path-loss RMSE (dB)	-3.474	[-3.896, -3.118]	<0.001	11/11	yes
Link transfer (ours) vs Tile labels (oracle)	path-loss RMSE (dB)	-2.316	[-2.610, -2.075]	<0.001	11/11	yes
Link + 10 tiles (ours) vs Link transfer (ours)	path-loss RMSE (dB)	+0.134	[+0.112, +0.156]	<0.001	0/11	yes
Link + 10 designed tiles (ours) vs Link + 10 tiles (ours)	path-loss RMSE (dB)	+0.061	[-0.008, +0.146]	0.278	5/11	no
Link + 50 designed tiles (ours) vs Link transfer (ours)	path-loss RMSE (dB)	+0.116	[+0.054, +0.187]	0.007	1/11	no
GBDT vs Source median	serving site	+0.0	[+0.0, +0.0]	1.000	0/11	no
Link transfer (ours) vs GBDT	serving site	+18.7	[+15.1, +22.6]	<0.001	11/11	yes
Link transfer (ours) vs Tile labels (oracle)	serving site	+18.7	[+15.3, +22.7]	<0.001	11/11	yes
Link + 10 tiles (ours) vs Link transfer (ours)	serving site	-2.9	[-4.5, -1.7]	<0.001	0/11	yes
Link + 10 designed tiles (ours) vs Link + 10 tiles (ours)	serving site	-0.2	[-1.4, +1.0]	0.765	6/11	no
Link + 50 designed tiles (ours) vs Link transfer (ours)	serving site	-1.8	[-4.1, -0.2]	0.067	4/11	no
GBDT vs Source median	SINR MAE (dB)	+0.258	[+0.156, +0.370]	<0.001	0/11	yes
Link transfer (ours) vs GBDT	SINR MAE (dB)	-2.069	[-2.448, -1.745]	<0.001	11/11	yes
Link transfer (ours) vs Tile labels (oracle)	SINR MAE (dB)	-2.198	[-2.516, -1.907]	<0.001	11/11	yes
Link + 10 tiles (ours) vs Link transfer (ours)	SINR MAE (dB)	+0.153	[+0.099, +0.213]	<0.001	0/11	yes
Link + 10 designed tiles (ours) vs Link + 10 tiles (ours)	SINR MAE (dB)	+0.136	[+0.059, +0.218]	0.014	3/11	no
Link + 50 designed tiles (ours) vs Link transfer (ours)	SINR MAE (dB)	+0.124	[+0.005, +0.266]	0.240	5/11	no
GBDT vs Source median	coverage agreement	+0.4	[-0.4, +1.5]	0.898	5/11	no
Link transfer (ours) vs GBDT	coverage agreement	+2.1	[+1.3, +2.9]	<0.001	11/11	yes
Link transfer (ours) vs Tile labels (oracle)	coverage agreement	+0.8	[+0.3, +1.2]	0.002	10/11	yes
Link + 10 tiles (ours) vs Link transfer (ours)	coverage agreement	-0.1	[-0.2, +0.1]	0.413	4/11	no
Link + 10 designed tiles (ours) vs Link + 10 tiles (ours)	coverage agreement	-0.2	[-0.4, -0.1]	0.014	2/11	no
Link + 50 designed tiles (ours) vs Link transfer (ours)	coverage agreement	-0.3	[-0.5, -0.1]	0.054	2/11	no
GBDT vs Source median	regret, $K=5$	-11.5	[-19.1, -3.8]	0.019	10/11	no
Link transfer (ours) vs GBDT	regret, $K=5$	-20.1	[-27.0, -14.1]	<0.001	11/11	yes
Link transfer (ours) vs Tile labels (oracle)	regret, $K=5$	-8.8	[-11.6, -6.1]	<0.001	11/11	yes
Link + 10 tiles (ours) vs Link transfer (ours)	regret, $K=5$	+0.0	[+0.0, +0.1]	0.312	4/11	no
Link + 10 designed tiles (ours) vs Link + 10 tiles (ours)	regret, $K=5$	+0.4	[+0.0, +1.1]	0.109	1/11	no
Link + 50 designed tiles (ours) vs Link transfer (ours)	regret, $K=5$	+0.0	[+0.0, +0.1]	0.250	0/11	no